Project Proposal: MS&E 342

Title: Reward-Conditioned Reverse Diffusion for Portfolio Optimization
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Course: MS&E 342 — Stochastic Systems and Learning Theory with Applications in Finance
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Abstract

We propose a framework for fine-tuning a score-based diffusion model's reverse stochastic differential equation (SDE) drift under a portfolio-utility reward, using the KL-penalized stochastic control theory developed in this course. The key observation is that existing work treats the diffusion model as a frozen scenario generator with no feedback from the downstream portfolio objective; and the stochastic control theory for reward fine-tuning of diffusion models (Han, Razaviyayn & Xu, ICML 2025) has never been applied to financial decision-making. We bridge these two threads, derive the HJB optimality condition for the fine-tuning problem, introduce optimal transport (OT) calibration as a distributional correction step, and validate the full pipeline empirically on S&P 500 sector ETF data.

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1. Motivation

1.1 The gap in the existing literature

Score-based diffusion models have recently been applied to financial scenario generation and portfolio optimization (Aghapour, Bayraktar & Yuan, NeurIPS 2025; Gao, Zha & Zhou, 2025). In these works the diffusion model is trained to approximate the historical return distribution and scenarios are sampled from it unconditionally. The portfolio optimizer then uses these scenarios as inputs. **The diffusion model and the portfolio objective are completely decoupled**: the generative model is never updated based on whether the scenarios it produces are useful for downstream decisions.

Concurrently, a rigorous stochastic control framework for reward fine-tuning of diffusion models has been developed (Han, Razaviyayn & Xu, ICML 2025; Domingo-Enrich et al., ICLR 2025). These works derive HJB equations, prove convergence of policy iteration, and show that fine-tuning the reverse SDE drift under a KL budget is well-posed. **Neither paper applies this theory to finance or to portfolio utility rewards.**

Our contribution is technically distinct from Aghapour, Bayraktar & Yuan (NeurIPS 2025): they model asset prices as a *diffusion-generated continuous-time process* and solve a dynamic HJB for the portfolio weight over the full price path, with state space (t, price_t) and a path-dependent objective. Our setting is categorically different — the diffusion model is a *scenario generator* for a terminal return vector, and we fine-tune the generator's reverse SDE drift rather than the portfolio weight. The two problems have different state spaces, different HJB equations, and different solution methods.

The bridge between the two threads — fine-tuning a financial scenario-generating diffusion model toward a terminal portfolio utility using the stochastic control machinery of Han/Razaviyayn/Xu — is the contribution of this project.

1.2 Why the decoupling matters

A scenario generator optimised for generic statistical fidelity (e.g., denoising score matching loss) does not necessarily produce scenarios that are decision-relevant. Our preliminary experiments make this concrete: a VP-SDE score model trained on 10 sector ETFs generates scenarios with approximately half the historical volatility, causing mean-variance optimisation to allocate 100% of the portfolio to a single asset. The Bures-Wasserstein distance between the generated and real return distributions is $W_2 = 0.028$, and the optimal transport map reveals that the model systematically underestimates cross-asset covariance. A scenario generator that is fine-tuned to be useful for portfolio decisions — while remaining close to the real distribution via a KL constraint — should correct this and produce better out-of-sample risk-adjusted performance.

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2. Mathematical Framework

2.1 Forward VP-SDE and score function

We model the joint daily log-return vector $X_0 \in \mathbb{R}^d$ under the variance-preserving SDE:

$$dX_t = -\tfrac{\beta(t)}{2}\, X_t\, dt + \sqrt{\beta(t)}\, dW_t, \qquad \beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min})$$

with marginal $X_t \mid X_0 \sim \mathcal{N}(\alpha_t X_0,\, \sigma_t^2 I)$ where $\alpha_t = \exp(-\frac{1}{2}\int_0^t \beta)$, $\sigma_t^2 = 1 - \alpha_t^2$. A score network s_θ is trained via denoising score matching. By the time-reversal theorem (Lecture 2-3), generating a scenario reduces to integrating the reverse SDE:

$$dY_t = [-\tfrac{\beta(T-t)}{2}\, Y_t + \beta(T-t)\, s_\theta(Y_t, T-t)]dt + \sqrt{\beta(T-t)}\, d\tilde{W}_t, \qquad Y_0 \sim \mathcal{N}(0, I)$$

2.2 Optimal transport calibration

Before fine-tuning, we align the generated distribution with the real data via optimal transport. By Brenier's theorem, the unique W_2-optimal map between two absolutely continuous measures is $T^* = \nabla\varphi$ for a convex potential φ. Under the Gaussian approximation, this is the closed-form Bures-Wasserstein map:

$$T^*(x) = \mu_q + A(x - \mu_p), \qquad A = \Sigma_p^{-1/2}\big(\Sigma_p^{1/2}\Sigma_q\Sigma_p^{1/2}\big)^{1/2}\Sigma_p^{-1/2}$$

For the non-Gaussian case, we use the Sinkhorn algorithm (entropy-regularised discrete OT) to compute a sample-level transport plan and barycentric projection. Crucially, OT calibration is incorporated at the **score model training level** (Contribution 2): we augment the denoising score matching loss with a differentiable Sinkhorn penalty, so p_θ itself is trained to match p_{data} in Wasserstein distance. This ensures the KL reference in problem ($\star$) is the OT-corrected base model, not the original miscalibrated one. Applying OT only post-hoc as a sample transformation would create a reference measure inconsistency: the KL penalty would penalise departures from a distribution we have already decided is wrong.

2.3 Fine-tuning as KL-penalized stochastic control

We introduce a control perturbation $u_\phi(Y_t, t)$ to the reverse drift and solve:

$$\max_{\phi} \; \mathbb{E}_{p_\phi}[U(Y_T)] - \eta \cdot \mathrm{KL}(p_\phi \,\|\, p_\theta) \tag{$\star$}$$

where $U(r) = \frac{1}{2\lambda} r^\top \Sigma^{-1} r$ is the mean-variance utility under optimal Markowitz weights, and the controlled reverse SDE is:

$$dY_t = [-\tfrac{\beta}{2} Y_t + \beta\, s_\theta(Y_t, t) + u_\phi(Y_t, t)]dt + \sqrt{\beta}\, d\tilde{W}_t$$

By Girsanov's theorem (Lecture 7), the KL penalty equals the quadratic control cost:

$$\mathrm{KL}(p_\phi \,\|\, p_\theta) = \mathbb{E}\left[\int_0^T \frac{\|u_\phi(Y_t, t)\|^2}{2\,\beta(T-t)}\, dt\right]$$

This is precisely the entropy-regularised stochastic control structure from Lecture 7, §7.1–7.2. Substituting the KL expression, problem ($\star$) becomes:

$$\max_u \mathbb{E}\left[U(Y_T) - \eta \int_0^T \frac{\|u_t\|^2}{2\beta(T{-}t)}\, dt\right]$$

The running cost is $-\frac{\eta}{2\beta}\|u\|^2$. The Hamiltonian maximization $\sup_u\{u \cdot \nabla_y V - \frac{\eta}{2\beta}\|u\|^2\}$ yields first-order condition $\nabla_y V = \frac{\eta}{\beta} u$, so:

$$u^*(t, y) = \frac{\beta(T{-}t)}{\eta}\, \nabla_y V(t, y)$$

Substituting back, the supremum equals $\frac{\beta(T-t)}{2\eta}\|\nabla_y V\|^2$. The Hamilton-Jacobi-Bellman equation for ($\star$) is therefore:

$$\partial_t V + \mathcal{L}^{s_\theta} V + \frac{\beta(T{-}t)}{2\eta}\|\nabla_y V\|^2 = 0, \qquad V(T, y) = U(y) \tag{HJB}$$

The scalar η is the **distributional budget**: small η makes the running cost cheap, so u^* can be large (aggressive reward-seeking); large η makes u^* small, keeping p_ϕ close to p_θ (realism-preserving). This is the continuous-time analogue of RLHF described in Lecture 13 — the pre-trained diffusion model plays the role of the reference policy and the portfolio utility plays the role of the reward model.

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3. Proposed Contributions

Contribution 1 — Theoretical (small theorem). In the linear-Gaussian special case ($p_{\text{data}} = \mathcal{N}(\mu, \Sigma)$ with perfect score $s_\theta(y,t) = -(y - \alpha_t \mu)/\sigma_t^2$, quadratic utility $U(r) = r^\top \Sigma^{-1} r/(2\lambda)$), we derive the closed-form solution to (HJB) via a quadratic ansatz $V(t,y) = \frac{1}{2} y^\top Q(t) y + b(t)^\top y + c(t)$ with terminal condition $Q(T) = \Sigma^{-1}/\lambda$, $b(T) = 0$. The resulting ODEs for $Q(t), b(t)$ are tractable, and the optimal control at the terminal time is:

$$u^*(T, y) = \frac{\beta(T)}{\eta\lambda} \Sigma^{-1} y$$

This is a drift proportional to $\Sigma^{-1} y$ — the Markowitz direction of maximum Sharpe ratio — with magnitude controlled by the budget η: smaller η allows a larger tilt. The fine-tuned terminal distribution remains Gaussian with covariance inflated in the principal eigendirections of Σ^{-1}, giving an interpretable geometric characterisation of what fine-tuning does: it increases scenario dispersion in the directions most rewarded by the Markowitz objective.

Contribution 2 — OT calibration layer. Show that Gaussian OT reduces $W_2(\text{generated}, \text{real})$ by over 99% and that this directly recovers near-real portfolio weight vectors. Augment the score model training with a differentiable Sinkhorn loss term to address the root cause (underestimated variance) during training rather than post-hoc. This is a self-contained result that is independently useful.

Contribution 3 — Empirical: η-Sharpe frontier. On S&P 500 sector ETF data (yfinance, 2014–2021 training, 2022–2024 testing), run the full pipeline for $\eta \in \{0.05, 0.1, 0.5, 1.0, 5.0\}$ and compare:

Strategy	Tests
Equal-weight	Naive benchmark
Markowitz (plug-in)	Classical baseline
Wasserstein-robust MVO	Static robustness baseline (Blanchet et al. 2022)
Diffusion-Base $\to$ MVO	Generative model, no fine-tuning
Diffusion + OT $\to$ MVO	OT calibration only
Diffusion + OT + FT (η)	Full pipeline, 5 values of η

Primary metrics: Sharpe ratio, CVaR(95%), max drawdown, turnover. The 2022 rate shock (largest multi-asset drawdown in the test window) is the main stress test. The η-Sharpe frontier plot is the empirical core of the paper.

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4. Preliminary Results

Data: 10 S&P 500 sector ETFs, yfinance, 2014-2024.
Score model: 3-layer MLP, VP-SDE, 500 training epochs, DSM loss converges to 0.19.
OT: Gaussian map computed from training data moments.

OT calibration (key finding):

	W_2 to real data	XLK weight (MVO)
Historical data	—	35.8%
Generated (raw)	0.0277	100%
After Gaussian OT	0.0002	36.1%

The base model severely underestimates volatility, causing MVO to concentrate entirely in one asset. The Gaussian OT map recovers near-real portfolio weights. *Note:* $W_2 = 0.0002$ *after Gaussian OT is near-zero by construction (the map minimises* W_2 *between Gaussian approximations); the meaningful diagnostic is the portfolio weight recovery, which is validated out-of-sample in the backtest.*

Portfolio backtest — full pipeline (test 2022–2024, monthly rebalancing):

Strategy	Ann. Return	Sharpe	95% CI Sharpe	Max Drawdown
Equal-weight	5.6%	0.36	[−0.76, +1.46]	−18.4%
Markowitz	9.0%	**0.52**	[−0.60, +1.57]	−17.1%
Wasserstein-Robust	5.8%	0.38	[−0.74, +1.48]	−17.9%
Diffusion-Base	11.5%	0.45	[−0.66, +1.55]	−33.1%
Diffusion + OT	6.1%	0.38	[−0.72, +1.50]	−22.3%
Diffusion + FT ($\eta^* = 0.1$)	10.2%	0.49	[−0.63, +1.60]	−29.2%
Diffusion + OT + FT ($\eta^* = 0.1$)	8.8%	0.47	[−0.69, +1.57]	**−26.4%**

The η-sweep over $\{0.05, 0.1, 0.5, 1.0, 5.0\}$ finds a peak at $\eta^* = 0.1$ for both pipelines. Key findings: (1) Fine-tuning alone at $\eta^* = 0.1$ recovers the highest Sharpe among diffusion strategies (0.49 vs. 0.45 base) with reduced drawdown (−29% vs. −33%). (2) OT calibration reduces drawdown the most in absolute terms (−22% vs. −33%), at the cost of some Sharpe. (3) The combined pipeline (OT + FT) achieves a balanced trade-off (0.47 Sharpe, −26% drawdown). Bootstrap 95% CIs on Sharpe are ±0.6–0.7 for the 3-year test window, so no individual result is statistically significant; the consistent pattern across the η-frontier is the interpretable finding.

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5. Course Connections

The project uses material from all three parts of the course:

Course content	Role in project
Fokker-Planck equation (Lec 2)	Governs density evolution under VP-SDE
Time-reversal theorem (Lec 2-3)	Reverse SDE for scenario generation
Denoising score matching (Lec 2-3)	Score network training objective
Tweedie's formula (Lec 2-3)	Denoised return estimate $\mathbb{E}[X_0 \mid X_t]$
HJB equation (Lec 7)	Optimality condition for fine-tuned control u^*
Entropy-regularised HJB (Lec 7, §7.1–7.2)	Exact structure of problem ($\star$)

Course content	Role in project
Girsanov / KL as quadratic cost (Lec 7–8)	$\mathrm{KL} = \mathbb{E}[\int \|u\|^2/2\beta\, dt]$
RLHF as stochastic control (Lec 13)	Fine-tuning formulation and policy iteration

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6. Timeline

	Milestone
Week 1 (done)	Data pipeline, VP-SDE score model, OT calibration, baseline backtest
Week 2	Retrain score model (2000 epochs + Sinkhorn augmentation); verify stylized facts
Week 3	η cross-validation sweep; full fine-tuning runs
Week 4	CVaR reward variant; η-Sharpe frontier plots
Week 5	Closed-form linear-Gaussian theory section
Week 6 (Jun 12)	Final write-up and submission

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7. References

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